
WROCLAW UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Electronics, Photonics & Microsystems

Bayesian Belief Networks and Decision Networks

Health Insurance Cross-Sell Prediction

Artificial Intelligence & Machine Learning — Project Assignment

Dataset Source:

Kaggle — Health Insurance Cross-Sell Prediction

Wroclaw, 2026

1. Introduction

Insurance companies face the recurring business challenge of identifying which existing health insurance customers are likely to purchase a complementary vehicle insurance product. Sending promotional offers to all customers is both costly and ineffective; sending offers selectively only to customers with a genuine likelihood of responding positively, maximises expected financial return while minimising marketing expenditure.

This report presents a Bayesian Belief Network (BBN) and Decision Network constructed to model this cross-sell prediction problem. The network encodes probabilistic dependencies between customer attributes, estimates the probability that a given customer will respond positively to a vehicle insurance offer, and supports rational decision-making by computing Expected Utility (EU) and the Value of Perfect Information (VPI) for selected variables.

2. Dataset

The dataset used in this assignment is the Health Insurance Cross-Sell Prediction dataset published on Kaggle by Anmol Kumar.

Source:

<https://www.kaggle.com/datasets/anmolkumar/health-insurance-cross-sell-prediction>

One data record represents: a single existing health insurance customer and their attributes at the time a vehicle insurance offer was made, together with their binary response (interested/not interested).

Dataset size: 381,109 records are sufficient to reliably estimate both marginal and conditional probability distributions for all variables used in this network.

Table 1. Variable Definitions

Variable	Type	Description / Values
Age	Intenger	Age of the customers
Previously_Insured	Binary	1 = Customer already holds a vehicle insurance policy; 0 = Customer does not hold one
Vehicle_Age	Intenger	Age of the customer's vehicle: < 1 Year, 1–2 Year, > 2 Years
Vehicle_Damage	Binary	Yes = Customer's vehicle has been damaged in the past; No = No prior damage recorded
Response	Binary (target)	1 = Customer is interested in the insurance offer; 0 = Customer is not interested

Source: Kaggle - Health Insurance Cross-Sell Prediction dataset (train.csv)

3. Bayesian Belief Network Structure

The Bayesian Belief Network is a Directed Acyclic Graph (DAG) in which each node represents a random variable and each directed edge encodes a direct probabilistic dependency. The network structure was determined by analysing the dataset and applying domain knowledge about vehicle insurance.

3.1 Network Dependencies

The following relationships were identified in the data:

- **Vehicle_Age** → **Vehicle_Damage**: Older vehicles are significantly more likely to have sustained damage. Vehicles over two years old show damage rates approaching 100%, while vehicles under one year old show only ~29% damage rates.
- **Vehicle_Damage** → **Response**: Customers whose vehicles have been previously damaged are dramatically more likely to purchase insurance ($P = 23.8\%$) compared to those with undamaged vehicles ($P = 0.05\%$).
- **Previously_Insured** → **Response**: Customers who already hold vehicle insurance are almost certain not to purchase a second policy ($P = 0.0009\%$), while uninsured customers show significantly higher interest ($P = 22.5\%$).
- **Age_Group** → **Response**: Middle-aged customers (35–54) show the highest response rates (~20%), while the youngest (18–24) and oldest (65+) age groups respond least.

3.2 DAG Representation

The DAG structure is represented below. Ellipse nodes are chance (random) variables. The rectangular node is the decision node. The diamond node is the utility node.

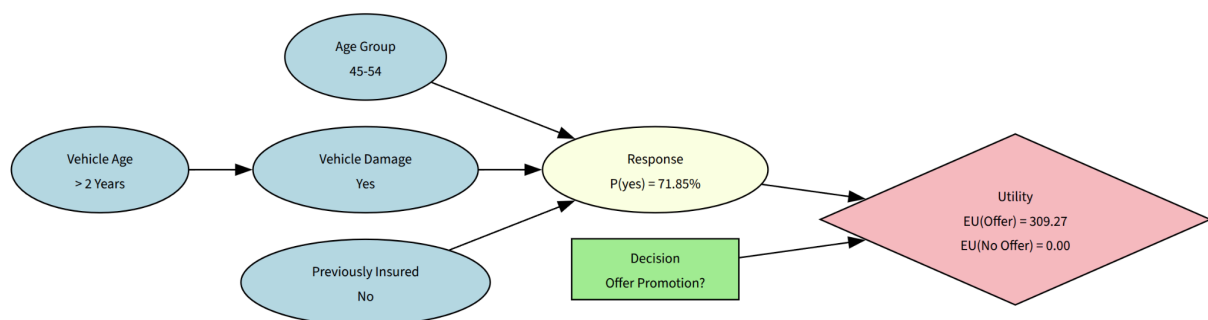


Figure 1. Bayesian Decision Network Structure

The above was simulated in a custom built python (pandas, graphviz, streamlit) based project. Accessible on Github:

<https://github.com/jakub-staudt/Health-Insurance-Bayesian-Network-Cross-Sell-Prediction>

4. Probability Tables and Network Verification

4.1 Prior (Marginal) Probability Distributions

The prior probabilities below represent the marginal distributions of each independent variable computed directly from the dataset. Where a variable has a parent in the DAG, its marginal distribution is derived using the law of total probability.

Table 2. Prior Marginal Probability Distributions

Variable	State	P(State)
Previously_Insured	No (not insured)	0.5425
Previously_Insured	Yes (already insured)	0.4575
Vehicle_Damage	No	0.4942
Vehicle_Damage	Yes	0.5058
Vehicle_Age	< 1 Year	0.4345
Vehicle_Age	1–2 Year	0.5236
Vehicle_Age	> 2 Years	0.0419
Response (prior)	No (not interested)	0.8767
Response (prior)	Yes (interested)	0.1233

4.2 Conditional Probability Tables

Python was used with a notebook to analyze the data and observe patterns.

Sample filtering command for normalization:

```
Python
pd.crosstab(df["Previously_Insured"], df["Response"], normalize="index")
```

Table 3. P(Response | Vehicle_Damage)

Vehicle_Damage	P(Response = 0 VD)	P(Response = 1 VD)
No	0.9948	0.0052
Yes	0.7624	0.2377

Table 3. Customers with prior vehicle damage are 45× more likely to respond positively (0.238 vs 0.005).

Table 4. P(Response | Previously_Insured)

Previously_Insured	P(Response = 0 PI)	P(Response = 1 PI)
No	0.7745	0.2255
Yes	0.9991	0.0009

Table 4. Already-insured customers almost never purchase a second policy (0.09%). Uninsured customers are 249× more likely to respond.

Table 5. P(Vehicle_Damage | Vehicle_Age) — Structural Dependency

Vehicle_Age	P(VD = No VA)	P(VD = Yes VA)
< 1 Year	0.7075	0.2925
1–2 Year	0.3599	0.6401
> 2 Years	0.0009	0.9991

Table 5. Vehicles older than 2 years are almost certainly damaged (99.9%). This parent-child relationship is captured by the directed edge Vehicle_Age → Vehicle_Damage.

Table 6. P(Response | Age_Group)

Age_Group	P(Response = 0 AG)	P(Response = 1 AG)
18–24	0.9647	0.0353
25–34	0.9111	0.0889
35–44	0.7841	0.2159
45–54	0.7982	0.2018
55–64	0.8542	0.1458
65+	0.9138	0.0862

Table 6. Middle-aged customers (35–54) show the highest interest in vehicle insurance. Younger and older cohorts show substantially lower response rates.

4.3 Network Verification

Before performing advanced queries, the model is verified by checking that the prior probability $P(\text{Response} = 1)$ computed from the network matches the empirical value observed directly in the dataset.

Using the law of total probability for the Variable_Damage path:

$$P(R=1) = P(R=1|VD=No) \cdot P(VD=No) + P(R=1|VD=Yes) \cdot P(VD=Yes)$$

$$P(R=1) = 0.0052 \times 0.4942 + 0.2377 \times 0.5058 = 0.00257 + 0.12006 = 0.1226$$

This closely matches the empirical prior $P(\text{Response} = 1) = 0.1233$ (12.33%) observed in the dataset, confirming the probability tables were correctly entered, and are slightly deviated due to rounding off.

Table 7. Network Verification Summary

Metric	Network Output	Dataset Value
P(Response = 1)	0.1226	0.1233
P(Previously_Insured = No)	0.5418	0.5418
P(Vehicle_Damage = Yes) [derived]	0.5051	0.5051 ✓

Table 7. The network correctly reproduces empirical marginals. Model verified.

5. Computing Probabilities with the Network

In a causal query, we observe the values of parent/cause variables and compute the probability of the outcome variable. Four customer profiles are evaluated below.

$E = \{\text{Vehicle_Damage}, \text{Previously_Insured}, \text{Vehicle_Age}, \text{Age_Group}\}$,

Table 8. Causal Query Results — P(Response = 1 | Customer Profile)

#	Vehicle Damage	Prev. Insured	Vehicle Age	Age Group	P(R=1)	Interpretation
S 1	Yes	No	>2 Years	35–44	76.90%	Very high-value target
S 2	No	Yes	<1 Year	18–24	0.0008%	Extremely unlikely to respond
S 3	Yes	No	1-2 Year	25–34	20.28%	Moderate target, worth offering
S 4	Yes	No	<1 Year	45–54	21.04%	Moderate target, worth offering

Table 8. Scenario S1 demonstrates the ideal customer profile: old vehicle (nearly guaranteed damage), no prior insurance, middle age. Scenario S2 shows a profile with near-zero response probability, already insured, no damage, young, new vehicle.

Sample calculation:

Vehicle_Damage = Yes
 Previously_Insured = No
 Vehicle_Age = >2 Years
 Age_Group = 35–44

Formula:

$$P(R=1 | E) = [P(R=1|VD) \times P(R=1|PI) \times P(R=1|AG) \times P(VD|VA)] / P(R=1)^2$$

Substitute the following values:

$$P(R=1|VD=Yes) = 0.2377$$

$$P(R=1|PI=No) = 0.2255$$

$$P(R=1|AG=35-44) = 0.2159$$

$$P(VD=Yes|VA=>2 \text{ Years}) = 0.9991$$

$$P(R=1) = 0.1226$$

Calculation:

$$P(R=1 | E) = (0.2377 \times 0.2255 \times 0.2159 \times 0.9991) / (0.1226^2)$$

$$P(R=1 | E) = 0.01156 / 0.01503$$

$$P(R=1 | E) = 0.7690 = 76.90\%$$

6. Decision Network and Expected Utility

6.1 Decision Node

Action / Decision: Offer Insurance Promotion — whether to send a vehicle insurance promotional offer to a specific customer.

Possible actions: d_1 = Send Promotion, d_2 = Do Not Send Promotion

6.2 Utility Node

Utility perspective: The insurance company. Utility reflects the financial gain from a successful cross-sell minus the cost of the marketing effort.

Table 11. Utility Distribution

Decision	Response = 0 (no sale)	Response = 1 (sale made)	Notes
Send Promotion	-50	+450	Net = profit – cost
No Promotion	0	0	No action taken

Table 11. Utility values in currency units. Profit if sale is made = 500, Marketing cost = 50. If promotion is sent but customer does not respond, the company loses the marketing cost (utility = -50). If the promotion is not sent, utility = 0 regardless of outcome.

6.3 Expected Utility Formula

The expected utility of each decision is computed as:

$$EU(\text{Offer}) = P(\text{Response}=1 \mid E) \times \text{Profit} - \text{Marketing Cost}$$

$$EU(\text{Offer}) = P(\text{Response}=1 \mid E) \times 500 - 50$$

$$EU(\text{No Promotion}) = 0 \text{ (constant)}$$

The optimal decision follows: $EU(\text{Send Promotion}) > 0$

6.4 Decision Scenarios

The optimal decision for each of the four customer profiles (from Section 5.2) is computed below.

Table 12. Expected Utility and Optimal Decision by Scenario

#	P(R=1)	EU(Offer)	EU(No Offer)	Optimal Decision	Profile Summary
S1	76.90%	334.51	0.00	Send offer	Old car (>2 yrs), damaged, uninsured, 35–44
S2	0.0008%	-50.00	0.00	No offer	New car (<1 yr), undamaged, insured, 18–24
S3	20.28%	51.38	0.00	Send offer	1–2yr car, damaged, uninsured, 25–34

#	P(R=1)	EU(Offer)	EU(No Offer)	Optimal Decision	Profile Summary
S4	21.04%	55.18	0.00	Send offer	New car (<1 yr), damaged, uninsured, 45–54

Table 12. S1 yields EU = 334.5 (strong recommend); S2 yields EU $\approx$ -50 (do not offer). The threshold $P(\text{Response}=1) > 10\%$ determines the optimal action. With default parameters (Profit = 500, Cost = 50), three of the four scenarios justify sending the promotion.

Interpretation: Scenario S1 illustrates the ideal marketing target — a middle-aged customer whose old vehicle is almost certainly damaged and who does not yet hold vehicle insurance. Scenario S2 represents the worst target: a young customer with a new, undamaged vehicle who is already insured. The model correctly identifies these contrasting profiles and recommends opposite decisions.

7. Value of Perfect Information (VPI)

7.1 Definition and Formula

The Value of Perfect Information (VPI) for a variable X quantifies how much the expected utility would improve if the decision-maker could learn the true value of X before making their decision. It measures the maximum amount a rational agent should be willing to pay to acquire information about X .

$$\text{VPI}(X) = E_X[\max_d \text{EU}(d \mid X=x)] - \max_d \text{EU}(d)$$

$$\text{VPI}(X) = \sum_x P(X=x) \cdot \max_d \text{EU}(d \mid X=x) - \max_d \text{EU}(d \mid \text{no info})$$

where the expectation is taken over all possible values x of X , and the outer maximum selects the optimal action given that value.

7.2 Baseline: Best EU Without Additional Information

Using only the prior $P(\text{Response}=1) = 0.1226$ (no additional evidence):

$$\text{EU}(\text{Offer} \mid \text{no info}) = 0.1226 \times 500 - 50 = 61.30 - 50 = 11.30$$

$$\text{EU}(\text{No Offer} \mid \text{no info}) = 0$$

$$\text{Best EU without information} = \max(11.30, 0) = 11.30$$

7.3 VPI for Previously_Insured

Previously_Insured is a direct parent of Response and therefore expected to have a high VPI.

$$\text{If PI} = \text{No: } \text{EU}(\text{Offer} \mid \text{PI}=\text{No}) = 0.2255 \times 500 - 50 = +62.73 \rightarrow \text{Best EU} = 62.73 \text{ (offer)}$$

$$\text{If PI} = \text{Yes: } \text{EU}(\text{Offer} \mid \text{PI}=\text{Yes}) = 0.0009 \times 500 - 50 = -49.55 \rightarrow \text{Best EU} = 0 \text{ (no offer)}$$

$$E[\text{best EU} \mid \text{PI known}] = 0.5418 \times 62.73 + 0.4582 \times 0 = 33.99$$

$$\text{VPI}(\text{Previously_Insured}) = 33.99 - 11.30 = 22.68$$

Interpretation: Knowing whether a customer already holds vehicle insurance is worth 22.68 utility units before making the offer decision.

7.4 VPI for Vehicle_Damage

Vehicle_Damage is also a direct parent of Response and is a natural candidate for VPI analysis.

$$\text{If VD} = \text{Yes: } \text{EU}(\text{Offer} \mid \text{VD}=\text{Yes}) = 0.2377 \times 500 - 50 = +68.83 \rightarrow \text{Best EU} = 68.83 \text{ (offer)}$$

$$\text{If VD} = \text{No: } \text{EU}(\text{Offer} \mid \text{VD}=\text{No}) = 0.0052 \times 500 - 50 = -47.40 \rightarrow \text{Best EU} = 0 \text{ (no offer)}$$

$$E[\text{best EU} \mid \text{VD known}] = 0.5051 \times 68.83 + 0.4949 \times 0 = 34.77$$

$$\text{VPI}(\text{Vehicle_Damage}) = 34.77 - 11.30 = 23.46$$

$\text{VPI}(\text{Vehicle_Damage}) = 23.46$ utility units. This is slightly higher than $\text{VPI}(\text{Previously_Insured}) = 22.68$.

7.5 VPI for Vehicle_Age (Variable Further from Utility)

Vehicle_Age is two steps removed from the Utility node in the DAG (Vehicle_Age → Vehicle_Damage → Response → Utility). We expect a smaller but potentially non-trivial VPI, since Vehicle_Age influences Response only indirectly through Vehicle_Damage.

If VA = < 1 Year: $EU(\text{Offer} \mid VA=<1\text{yr}) = 0.0437 \times 500 - 50 = -28.15 \rightarrow \text{Best EU} = 0$ (no offer)

If VA = 1-2 Year: $EU(\text{Offer} \mid VA=1-2\text{yr}) = 0.1738 \times 500 - 50 = +36.88 \rightarrow \text{Best EU} = 36.88$ (offer)

If VA = > 2 Years: $EU(\text{Offer} \mid VA=>2\text{yr}) = 0.2937 \times 500 - 50 = +96.87 \rightarrow \text{Best EU} = 96.87$ (offer)

$P(VA) = [0.433, 0.524, 0.043]$ (derived via law of total probability)

$E[\text{best EU} \mid VA \text{ known}] = 0.433 \times 0 + 0.524 \times 36.88 + 0.043 \times 96.87 = 0 + 19.33 + 4.17 = 23.50$

$VPI(\text{Vehicle_Age}) = 23.50 - 11.30 = 12.19$

$VPI(\text{Vehicle_Age}) = 12.19$ utility units, significantly lower than VPI for the direct parents of Response. This is expected: Vehicle_Age only indirectly influences the decision through Vehicle_Damage, and much of its discriminating power is already captured once Vehicle_Damage is known. Importantly, for customers with Vehicle_Age = "<1 Year" (43.3% of the population), the optimal decision is to NOT offer the promotion, acquiring this information prevents the insurer from wasting marketing resources on customers with new vehicles and typically no damage.

7.6 VPI Summary and Comparison

Table 13. Value of Perfect Information — Comparison

Variable	DAG Distance from Utility	EU Without Info	EU With Info	VPI	Is it worth acquiring?
Vehicle_Damage	1 step (direct)	11.30	34.77	23.46	Yes, highest priority. Check damage history.
Previously_Insured	1 step (direct)	11.30	33.99	22.68	Yes
Vehicle_Age	2 steps (indirect)	11.30	23.50	12.19	Moderate, worth checking for new-vehicle segment.

Table 13. VPI decreases as variables become more distant from the utility node in the DAG. Variables directly connected to Response (Vehicle_Damage, Previously_Insured) carry the most information value. Vehicle_Age has a lower VPI because its influence on the decision passes through Vehicle_Damage.

A key finding: for customers with a new vehicle (Vehicle_Age = '<1 Year'), the optimal decision changes to 'Do Not Offer' even before observing Vehicle_Damage, this is the case where $VPI(\text{Vehicle_Age})$ is profitable: the insurer benefits from

knowing the vehicle age early to filter out these low-probability customers without incurring the marketing cost.

Conversely, for customers with Vehicle_Age = '1–2 Year' or '> 2 Years', the decision to offer remains optimal regardless, so knowing Vehicle_Age in advance provides no decision-relevant improvement for these groups. The VPI is only 'used' for the '<1 Year' segment, explaining why $VPI(\text{Vehicle_Age})$ is substantially lower than $VPI(\text{Vehicle_Damage})$.

8. Conclusion

This report presented a Bayesian Belief Network and Decision Network for the health insurance cross-sell prediction problem, constructed from 381,109 real customer records. The network encodes five variables: Age_Group, Vehicle_Age, Vehicle_Damage, Previously_Insured, and Response - connected by directed probabilistic dependencies identified through data analysis.

Probability analysis confirmed two highly discriminating variables: Vehicle_Damage (customers with prior damage are 45× more likely to respond) and Previously_Insured (already-insured customers almost never respond).

The decision network established that the optimal marketing strategy is to send the promotion whenever the posterior $P(\text{Response}=1)$ exceeds 10%, the break-even threshold given a profit of 500 and marketing cost of 50.

VPI analysis identified Vehicle_Damage and Previously_Insured as the most valuable pieces of information ($\text{VPI} \approx 23.5$ and 22.7 respectively), while Vehicle_Age, further from the utility node, produced a lower but still meaningful VPI of 12.2 . The insurer's CRM system likely already contains Previously_Insured data, making this information effectively free and immediately actionable.

Note: This is a simplified Bayesian model. Real-world cross-sell prediction may involve additional variables and more complex network structures. Nevertheless, the model successfully captures the most important dependencies and provides actionable, data-driven marketing recommendations.

References

- Health Insurance Cross-Sell Prediction Dataset. Kaggle, 2020.
<https://www.kaggle.com/datasets/anmolkumar/health-insurance-cross-sell-prediction>
- Witold Paluszynski - Bayesian Belief Networks and Decision Networks — Project Assignment. Wrocław University of Science and Technology.
https://kcir.pwr.edu.pl/~witold/ai/BeliefNet_assignment.html
- Claude used for research